ANDROID PROGRAMMING PRACTICAL 09

Module 2 · Static and Dynamic Broadcast Receivers

Aim

Implement static and dynamic Broadcast Receivers.

Project name: BroadcastReceiverDemo

Requirements

Android Studio, Java, Android SDK, and an Android Virtual Device or compatible Android phone. Use the package name generated by your project and adjust imports/dependencies to the installed Android Studio template.

Procedure

1. Create a BroadcastReceiver subclass and override onReceive().

2. For a dynamic receiver, register it in an Activity using registerReceiver() with an IntentFilter, and unregister it at the matching lifecycle point.

3. For a static receiver, declare the receiver and appropriate intent filter in AndroidManifest.xml only for supported/appropriate broadcasts.

4. Use an app-defined broadcast action for a controlled demonstration where possible.

5. Send the matching broadcast Intent from a button or test component.

6. Run the app and verify onReceive() executes, using Logcat or Toast for feedback.

7. Check Android version restrictions and exported/permission requirements before testing manifest-declared receivers.

Implementation checklist

• Keep resource names lowercase and use unique view IDs.

• Make sure every referenced layout, menu, drawable, string, or Java class exists in the project.

• Build the project and resolve compile errors before running it.

• Use Logcat to diagnose runtime errors and permissions or manifest issues.

Expected result

The receiver processes the matching broadcast and logs or displays a response.

Testing / observations

Record the device/API level, the actions tested, the observed output, and any errors corrected. Attach a screenshot of the successful run to your practical record if required by your institution.

Conclusion

The stated Android concept was implemented and verified through the steps and tests above.